Arsalene Zbidi

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PROFESSIONAL SUMMARY

AI Engineer, PhD researcher, and AI team leader specializing in agentic AI systems and production ML. Proven track record reducing operational overhead by 70% and deployment cycles by 60%. Combining academic research in autonomous threat hunting with industry expertise in NLP, computer vision, and enterprise AI solutions for telecommunications, ERP, and cybersecurity sectors.

CORE SKILLS

AI Agents, Agentic AI, Multi-Agent Systems, LangChain, LangGraph, RAG, NLP, Computer Vision, YOLOv8, TensorFlow, PyTorch, OpenCV, MCP, Python, Java, FastAPI, PostgreSQL, MongoDB, Elasticsearch, Redis, Docker, Kubernetes, ERP, Chatbot Development, Conversational AI, Business Intelligence, DevOps, CI/CD

PROFESSIONAL EXPERIENCE

Head of AI | **Mar 2026 - Present**
Microzed Solution | *Megrine, Tunisia*

- Lead a team of 2 AI engineers researching, designing, and building AI agents integrated into the Zsolution ERP platform
- Architected an enterprise AI assistant chatbot unifying ERP data across BI, marketing, HR, dispatching, stock management, finance, and statistics, generating automated reports and on-demand insights
- Designed the assistant to surface proactive insights beyond raw ERP data - synthesizing new facts, patterns, and recommendations from cross-module data to support strategic decision-making
- Enabled natural language and voice-driven interaction with the ERP, allowing users to query data and trigger system actions conversationally

AI Engineer & Solution Architect | **Oct 2024 - Mar 2026**
Artificial Insight | *Tunis, Tunisia*

- Architected PowerSale AI platform processing 10TB+ telecom data daily, leading technical strategy for 5-person engineering team
- Pioneered Telco GPT conversational AI reducing analyst workload by 70% through LangGraph multi-agent orchestration
- Engineered KFF RAG System, a multi-tenant agentic platform for Kuwait Fire Force digitizing 1,000+ pages of training manuals with hybrid search (Qdrant semantic + Elasticsearch keyword), multi-modal OCR (DeepSeek, Tesseract) for Arabic text extraction, and context-aware retrieval via FalkorDB graph memory serving specialized agent workflows
- Developed MeetingMind AI pipeline converting audio meetings into actionable insights through real-time multilingual STT transcription, automated summary generation, and natural TTS audio reports, streamlining executive decision-making workflows
- Built modular backend infrastructure achieving sub-second queries through PostgreSQL, Redis caching, and hybrid chunking strategies, deployed with Llama 3.2 70B and GPT-OSS 20B via Ollama for on-premise LLM inference

Machine Learning Engineer Intern | **Feb 2024 - Sep 2024**
Artificial Insight | *Tunis, Tunisia*

- Delivered end-to-end vehicle recognition system reducing manual verification time by 80% for law enforcement applications
- Trained YOLOv8 models on 10,000+ annotated images achieving 95%+ detection accuracy at 30+ FPS on edge devices
- Built OCR pipeline (OpenCV, EasyOCR, TensorFlow CNN) achieving 92% accuracy in license plate recognition

- Implemented SORT tracking with MongoDB alerting, reducing false positives by 65% through unique vehicle identification

DevOps Engineer Intern | **Jan 2021 - Jun 2021**
Polina Group Holding | *Tunis, Tunisia*

- Spearheaded CI/CD transformation (GitLab CI, Jenkins, ArgoCD, Kubernetes) reducing deployment time by 60%
- Established monitoring framework (Grafana, Prometheus, SonarQube) improving system reliability by 45%

EDUCATION

PhD in Computer Science - Cybersecurity & AI | **2024 - Present**
Higher School of Communications of Tunis (SUP'COM) | *Tunis, Tunisia*

- Research Focus: Threat Hunting Using Agentic AI - Developing autonomous multi-agent systems for proactive cybersecurity threat detection and response

Engineering Degree in Computer Science - Data Science & AI | **2021 - 2024**
Central Polytechnic School of Tunis | *Tunis, Tunisia*

Bachelor in Computer Science | **2018 - 2021**
Higher Institute of Information and Communication Technologies | *Tunis, Tunisia*

TECHNICAL SKILLS

AI & ML: LangChain, LangGraph, RAG, Multi-Agent Systems, NLP, YOLOv8, TensorFlow, PyTorch, OpenCV, MCP

Development: Python, Java, FastAPI, Symfony, Angular, PHP

Data & Infrastructure: PostgreSQL, MongoDB, Elasticsearch, Redis, FalkorDB, Qdrant, Power BI, Docker, Kubernetes

DevOps: GitLab CI, Jenkins, ArgoCD, Prometheus, Grafana, SonarQube

Languages: English (Fluent), French (Fluent), Arabic (Native)

KEY PROJECTS & ACHIEVEMENTS

Multi-Agent Conversational AI | **LangGraph, MCP, Ollama**

- Built specialized agent architecture for autonomous task decomposition and collaborative problem-solving in production

ML Recommendation Engine | **PyTorch, TensorFlow**

- Developed clustering algorithm for behavioral analysis, increasing user engagement by 35%

Leadership

Captain of National Volleyball Team, Tunisia (2014-2018) | Captain of the Hammam-lif Senior Volleyball Team (2016-current)